CPU and decode

File: cpu.kicad_sch

Memory

File: memory.kicad_sch

I/O

File: io.kicad_sch

Power

File: power.kicad_sch

Memory map (24-bit, bank:offset)
  $00:0000-$00:5FFF   RAM (U5)
  $00:6000-$00:7FFF   I/O window, U7 '138, 8 x 1 KB slots
                      $6000 VIA (U8), $7000 ACIA (U9)
  $00:8000-$00:FFFF   ROM (U4)
  $01:0000-$07:FFFF   RAM (U5)
  $08:0000-$7F:FFFF   aliases of banks $00-$07
  $80:0000-$80:7FFF   video RAM (U6), aliased through $FF:FFFF

Naming: active-low signals are drawn with an overbar, e.g. $\overline{\text{IOSEL}}$, $\overline{\text{RAMCE}}$, R/$\overline{\text{W}}$.
The W65C datasheets write these with a B suffix (IRQB, RESB, RWB, CS2B);
decode.pld writes them with a leading '/' in its equations.

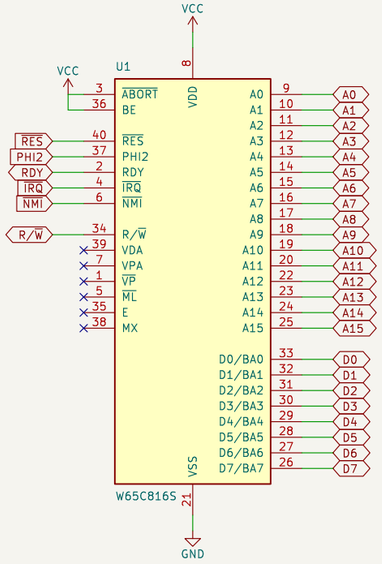
Active-low signals use KiCad overbar notation, e.g. $\overline{\text{IOSEL}}$
W65C816S breadboard computer - see README.md and pld/decode.pld
be6502

Sheet: /
File: be6502.kicad_sch

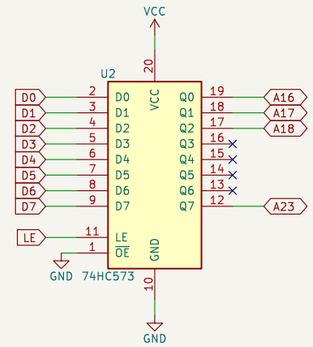
Title: be6502 - W65C816S breadboard computer

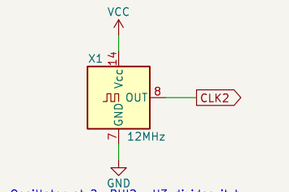
Size: A4	Date: 2026-10-16	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

W65C816S (65C816 in DIP-40).
D0–D7 carry the bank byte while PHI2 is low;
U2 captures it while LE is high (see NOTE 6 of decode.pld).
VDA, VPA, VP, ML, E and MX are unused.

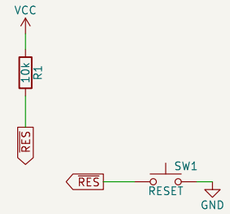


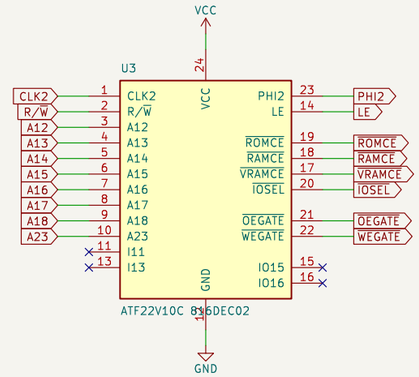
Bank byte latch. Q3–Q6 (A19–A22) are deliberately
not used: banks $08–$7F alias $00–$07 and $81–$FF
alias $80 (decode.pld NOTE 3).

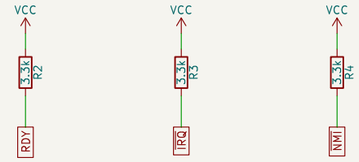




Oscillator at 2x PHI2. U3 divides it by
two to make PHI2 (bios.s: PHI2_HZ = 6 MHz).





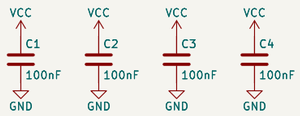


RDY is bidirectional on the '816 (WAI drives it low),
so it is pulled up rather than tied to VCC.
IRQ is shared: U9 (open drain) and U8 via D1.

U3 runs pld/decode.pld (816DEC02):

```
  PHI2.R    = /PHI2                 (CLK2 / 2)
  LE        = /PHI2 * /CLK2
  /ROMCE    = /A23*/A18*/A17*/A16*A15
  /RAMCE    = bank 0 $0000–$5FFF, or /A23 * (A16+A17+A18)
  /VRAMCE   = A23
  /IOSEL    = /A23*/A18*/A17*/A16*/A15*A14*A13
  /OEGATE   = PHI2 * RWB
  /WEGATE   = PHI2 * /RWB
```

A12 is wired but unused (NOTE 4). Pins 11, 13, 15, 16 spare.
Burn with power-down disabled (NOTE 5).



Decoupling: C1 U1, C2 U2, C3 U3, C4 X1 — one 100nF per package, close to the pins.

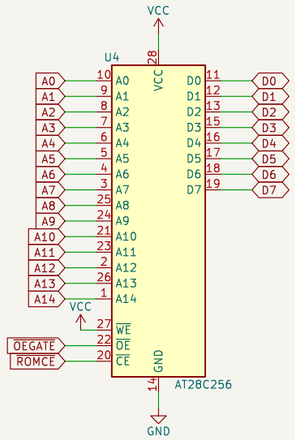
Active-low signals use KiCad overbar notation, e.g. $\overline{\text{IOSEL}}$
W65C816S breadboard computer — see README.md and pld/decode.pld
be6502

Sheet: /CPU and decode/
File: cpu.kicad_sch

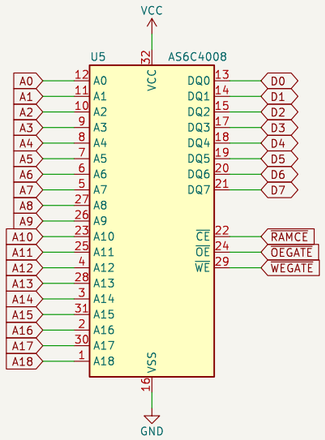
Title: CPU, clock, bank latch and address decode

Size: A3	Date: 2026-10-16	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/5

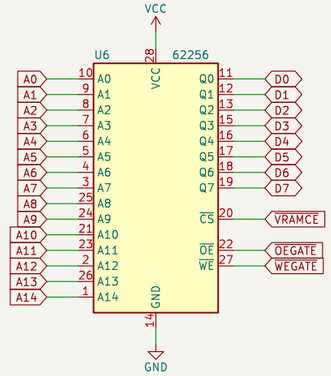
ROM $00:8000–$00:FFFF (32 KB)
WE tied high: not writable in circuit.



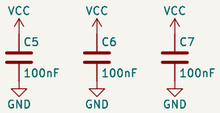
RAM 512 KB. A16–A18 come from the bank latch (U2).
$00:0000–$00:5FFF and $01:0000–$07:FFFF reachable.



Video RAM $80:0000–$80:7FFF (32 KB),
selected by A23 alone.



OEGATE and WEGATE are shared by all three memories; the chip enables
are mutually exclusive, so at most one device drives D0–D7 (decode.pld NOTE 1).
WEGATE includes PHI2, so no write can happen while D0–D7 carries the bank byte.

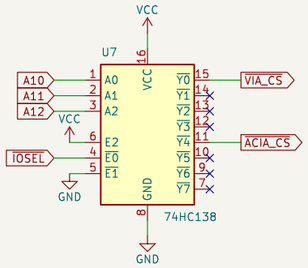


Decoupling: C5 U4, C6 U5, C7 U6.

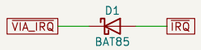
Active–low signals use KiCad overbar notation, e.g. IOSEL
W65C816S breadboard computer — see README.md and pld/decode.pld
be6502

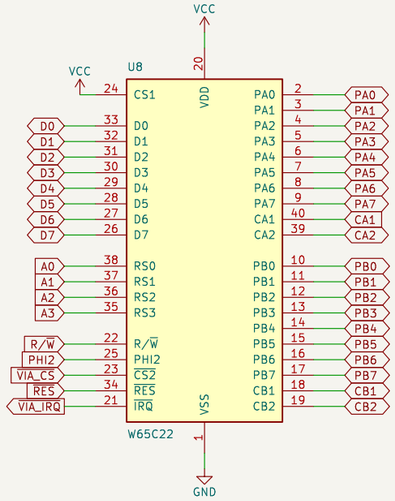
Sheet: /Memory/		
File: memory.kicad_sch		
Title: Memory: ROM, RAM, video RAM		
Size: A3	Date: 2026–10–16	**Rev: 1**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 3/5

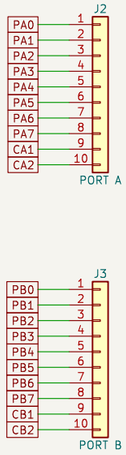
I/O window $00:6000–$00:7FFF, 8 x 1 KB slots on A12:A11:A10
  Y0 $6000 VIA   Y4 $7000 ACIA
  Y1 $6400 Y2 $6800 Y3 $6C00
  Y5 $7400 Y6 $7800 Y7 $7C00 (unallocated)
E1 (pin 5) is the reserved PHI2 hook: tie to PHI2
to qualify the selects later (decode.pld NOTE 2).
E2 (pin 6) is the active-high enable — it must be high.



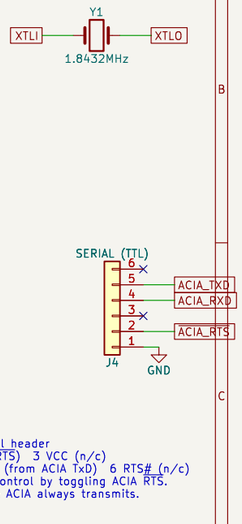
D1 lets a W65C22S (totem-pole IRQB) share IRQ with
the open-drain ACIA; also works with a W65C22N.
Use a Schottky (< 0.5 V drop). Pull-up R3 is on the CPU sheet.

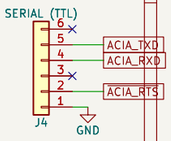




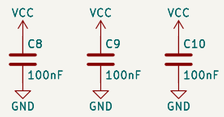


W65C22S ports have no current limiting: add series
resistors when driving LEDs or an HD44780.





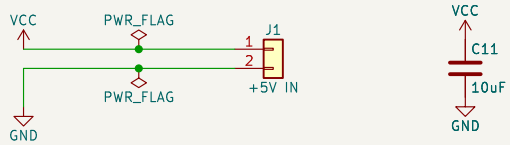
J4: FTDI-style 6-pin TTL serial header
  1 GND 2 CTS# (from ACIA RTS) 3 VCC (n/c)
  4 TXD (to ACIA RxD) 5 RXD (from ACIA TxD) 6 RTS# (n/c)
bios.s: 115200 8-N-1, flow control by toggling ACIA RTS.
CTS, DSR, DCD tied low so the ACIA always transmits.



Decoupling: C8 U7, C9 U8, C10 U9.

Active-low signals use KiCad overbar notation, e.g. IOSEL
W65C816S breadboard computer — see README.md and pld/decode.pld
be6502

Sheet: /I/O/		
File: io.kicad_sch		
Title: I/O: '138 slot decoder, VIA, ACIA		
Size: A3	Date: 2026-10-16	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5



Regulated +5 V in. C11 bulk capacitor at the supply entry.

Active-low signals use KiCad overbar notation, e.g. $\overline{\text{IOSEL}}$
W65C816S breadboard computer — see README.md and pld/decode.pld
be6502

Sheet: /Power/
File: power.kicad_sch

Title: Power

Size: A4	Date: 2026-10-16	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5